

Chain models and their effective Hamiltonians

M. Znojil

Institute of Nuclear Physics, Czechoslovak Academy of Sciences, 250 68 Řež, Czechoslovakia

(Received 1 August 1983; accepted for publication 9 March 1984)

We develop a new formalism suitable for diagonalization of Hamiltonians with $2t + 1$ nonzero diagonals (chain models). In a systematic perturbation-like way, we get the expansions of Green and/or wave functions. They are generated by solution of a sequence of "fixed-point" quadratic equations for $(t \times t)$ -dimensional matrices. For $t = 1$ and $t = 2$, this solution is feasible by elementary means, so that the respective tridiagonal and pentadiagonal chain models may be considered exactly solvable in this context. In an alternative first-order variational-perturbation formulation, the method may provide the simultaneous upper and lower energy bounds for any t .

PACS numbers: 03.65.Ge, 03.65.Db, 02.60.+y

I. INTRODUCTION

The concept of a strong interaction between neighboring "orbitals" has inspired the introduction and detailed study of the so-called chain models in solid-state physics,¹ in scattering or nuclear theory,² in the recent papers on the anharmonic oscillators,^{3,4} etc. These models are all characterized by the Schrödinger equation

$$H\psi = E\psi, \quad (1.1)$$

with some infinite-dimensional band-matrix Hamiltonian H .

By definition, the off-diagonal matrix elements of H are not small and the standard perturbation theory may converge only after a careful choice of the unperturbed operator H_0 .^{4,5} Moreover, there is a conflict between an easy diagonalization of this H_0 and its "realistic" character, i.e., between the simplicity of the perturbation expansions and their convergence. Therefore, the nonsmallness of the perturbation $H - H_0$ usually requires the nonperturbative treatment of H .

For the simplest tridiagonal Hamiltonians H , the nonperturbative techniques of their diagonalization¹ range from the analytic theory of continued fractions⁶ and analogies with the Mathieu functions to the general theory of moments⁷ and Padé approximants.⁸

Any generalization of this mathematics to band matrices with $2t + 1$ diagonals, $t > 1$, encounters immediately a number of specific methodical difficulties.^{3,4,9} Our present intention is to show that most of these difficulties disappear at least in the $t = 2$ case of five diagonals.

The material is organized as follows. In Sec. II, we review the "matrix continued fraction" (MCF) approach⁹ to diagonalization of the band matrices and emphasize that the MCF algorithm need not be employed in the numerical context only. Indeed, for the particular $t = 2$ anharmonic-oscillator (AHO) example,³ we are able to obtain also the explicit asymptotic form of the effective Hamiltonian (Appendix A) and/or to analyze its MCF convergence still by purely analytic means (Appendix B).

When applied to the general MCF quantity, this geometric "fixed-point" (FP) approach¹⁰ generates the natural FP approximants. In the chain-model context, they define algebraically the approximate effective Hamiltonians in a large model space (Sec. III).

In Sec. IV, a systematic FP treatment of the higher-order corrections is suggested. For the Schrödinger equation (1.1), this enables one to define exactly the effective Hamiltonian by an infinite series in an arbitrary model space. This general FP expansion resembles perturbation theory but its algebraic construction has a more complicated character (Appendix C), especially for $t > 1$.

The particular $t = 1$ (Sec. V A) and $t = 2$ (Sec. V B) versions of the FP expansions are exceptional and simple: Their construction remains fully elementary. This makes the pentadiagonal chain model "exactly solvable" and formally close to the current tridiagonal Hamiltonians.

II. BAND-MATRIX HAMILTONIANS AND THE MCF ALGORITHM

A. Effective Hamiltonians and bound states

Let us start by a reformulation of the MCF formalism initiated by Graffi and Grecchi^{3,9} and applicable to any Schrödinger-type eigenvalue problem:

$$\mathcal{H}(E)\psi = 0, \quad (2.1)$$

with the band or, equivalently,⁹ block-tridiagonal "Hamiltonian"

$$\mathcal{H}(E) = \begin{pmatrix} A_0 & B_0 & & \\ C_1 & A_1 & B_1 & \\ & C_2 & A_2 & B_2 \\ & & & \ddots \end{pmatrix}, \quad (2.2)$$

$$\dim A_0 = M_0 \geq 1, \quad \dim A_k = M_k = t \geq 1, \quad k = 1, 2, \dots$$

First, we introduce a model-space projector

$$P = P_{(M)} = |0\rangle\langle 0| + |1\rangle\langle 1| + \dots + |d-1\rangle\langle d-1|,$$

$$d = M_0 + M_1 + \dots + M_M.$$

In the spirit of Feshbach,¹¹ we may rewrite (2.1), i.e.,

$$P\mathcal{H}P\psi + P\mathcal{H}Q\psi = 0, \quad Q = 1 - P,$$

$$Q\mathcal{H}P\psi + Q\mathcal{H}Q\psi = 0,$$

in the equivalent form

$$\mathcal{H}_{(M)}^{\text{eff}} P_\psi = 0, \quad \mathcal{H}_{(M)}^{\text{eff}} = P\mathcal{H}P - P\mathcal{H}Q(1/Q\mathcal{H}Q)Q\mathcal{H}P, \quad (2.3)$$

$$Q_\psi = -(1/Q\mathcal{H}Q)Q\mathcal{H}P_\psi,$$

where, obviously, the complexity of solution concentrates in

the construction of the effective matrix $\mathcal{H}_{(M)}^{\text{eff}}$ itself.

In this formulation, the main MCF idea stems from the M -dependence of the exact effective Hamiltonians $\mathcal{H}_{(M)}^{\text{eff}}(E)$,

$$\mathcal{H}_{(0)}^{\text{eff}}(E) = G_0, \quad \mathcal{H}_{(1)}^{\text{eff}}(E) = \begin{pmatrix} A_0 & B_0 \\ C_1 & G_1 \end{pmatrix},$$

$$\mathcal{H}_{(2)}^{\text{eff}}(E) = \begin{pmatrix} A_0 & B_0 & & \\ C_1 & A_1 & B_1 & \\ & C_2 & G_2 & \end{pmatrix}, \dots \quad (2.4)$$

Indeed, when we choose the various dimensions $d = M_0 + tM$ of the model space, the different $(t \times t)$ -dimensional “effective” submatrices $G_0, G_1, \dots$ become related precisely by the MCF recurrences

$$G_k = A_k - B_k(1/G_{k+1})C_{k+1}, \quad k = 0, 1, \dots \quad (2.5)$$

These recurrent relations are to be initialized in accord with the standard variational technique [truncation of $\mathcal{H}(E)$], i.e., by $G_N = A_N$, $N \rightarrow \infty$. All the effective Hamiltonians (2.4) become exact in the MCF limit $N \rightarrow \infty$.

The dimension M_0 is variable. Its change may be used as a “regularization procedure” whenever we encounter the random zeros of $\det G_{k+1}$ at some $k = k_0 \geq 1$ in (2.5) (an example may be found, e.g., in Ref. 12). In the “regular” cases, we may fix $M_0 = 1$ and get the simplest MCF form of the secular equation

$$(A_0)_{11} - \sum_{i,j=1}^t (B_0)_{1i}(G_1^{-1})_{ij}(C_1)_{j1} = 0, \quad (2.6)$$

which is extremely suitable for a numerical search for the binding energies E . Quite formally, the inverse of the left-hand side expression (or $\det G_0$ in general) may be interpreted also as the MCF Green function.^{3,9}

Concerning the wave functions ψ , we may notice that their MCF form

$$\mathbf{X}_k = D_k D_{k-1} \dots D_1 \mathbf{X}_0, \quad D_k = -(1/G_k)C_k, \quad (2.7)$$

$$(\mathbf{X}_k)_m = \langle kt + m - 1 | \psi \rangle, \quad m = 1, 2, \dots, t, \quad k = 1, 2, \dots,$$

follows from (2.3) with $k = M \neq 0$. It must be accompanied by the remaining $M = 0$ requirement (2.3)

$$G_0 \mathbf{X}_0 = 0. \quad (2.8)$$

Then, the projections (2.7) satisfy (2.1) at the physical energies only.

Vice versa, any initialization G_0 and $\mathbf{X}_0$ of the respective definitions (2.5) and (2.7) leads to the formal solution $\langle n | \psi \rangle$ provided that (2.8) holds. Then, the requirement $\|\psi\| < \infty$

$$A_k = \begin{pmatrix} (2a_n - E)/\lambda + b_{n-1}^2 + a_n^2 + b_n^2 & (a_n + a_{n+1})b_n \\ (a_n + a_{n+1})b_n & (2a_{n+1} - E)/\lambda + b_n^2 + a_{n+1}^2 + b_{n+1}^2 \end{pmatrix},$$

$$B_k = C_{k+1}^T = \begin{pmatrix} b_n b_{n+1} & 0 \\ (a_{n+1} + a_{n+2})b_{n+1} & b_{n+1} b_{n+2} \end{pmatrix}, \quad k \geq 0. \quad (2.13)$$

Similarly, for any $t \geq 2$, the MCF numerical algorithm is easily implemented to produce the AHO energies.³

With $t < 0$, the singularly anharmonic Hamiltonians (2.9) may also be converted into the $(2|t| + 1)$ -diagonal matrices $\mathcal{H}(E)$: It suffices to use the nonorthogonal basis func-

must be imposed in a way illustrated again in Ref. 12 or in Sec. V A below (see alternative proof of Theorem 2).

B. Examples of the band-matrix Hamiltonians

Anharmonic oscillators (AHO) possess some interesting physical interpretations (e.g., in zero-dimensional field theory) and are often employed due to their technical simplicity.³⁻⁵ In the quantum-mechanical context, we may represent their Hamiltonians

$$H = -\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + r_2 + \lambda r^{2t},$$

$$\lambda > 0, \quad t = 2, 3, \dots, \quad (2.9)$$

in the harmonic-oscillator (HO) basis $|n\rangle$ and get the Schrödinger equation (1.1) or (2.1) in the band-matrix form:

$$0 = \sum_{n=m-t}^{m+t} [(2a_m - E)\delta_{mn} + \lambda \langle m | r^{2t} | n \rangle] \langle n | \psi \rangle, \quad (2.10)$$

$$a_m = 2m + l + \frac{3}{2}, \quad m = 0, 1, \dots$$

The matrix elements are easily obtainable since $\langle m | r^{2t} | n \rangle = (T^t)_{mn}$, where

$$T = \begin{pmatrix} a_0 & b_0 & & \\ b_0 & a_1 & b_1 & \\ & b_1 & a_2 & b_2 \\ & & & \ddots \end{pmatrix}, \quad b_m = [(m+1)(m+l+\frac{3}{2})]^{1/2},$$

with $l = -1, 0$ (parity in one dimension), $l = 0, 1, \dots$ (angular momentum in three dimensions), or $l > -\frac{1}{2}$ (noninteger parameter simulating the r^{-2} component in the potential¹²).

The trivial choice of $t = 1$ corresponds merely to the harmonic oscillator represented in a “wrong” basis. With

$$A_k = (2 + \lambda)a_k - E, \quad B_k = \lambda b_k, \quad t = 1, \quad k \geq 0, \quad (2.11)$$

this simplest tridiagonal chain model may be used as a methodical guide. In the present MCF context, we may immediately notice that the HO formula (2.5),

$$G_k = (2 + \lambda)a_k - E - \frac{\lambda^2 b_k^2}{(2 + \lambda)a_{k+1} - E - \dots}, \quad (2.12)$$

represents a classical continued fraction. It converges if and only if $\lambda > -1$, i.e., in all cases of interest.

In the simplest nontrivial $t = 2$ example, the matrix $\mathcal{H}(E)$ may be given the form (2.2) with $2k + M_0 - 2 = n$ and

tions $\langle r | n \rangle r^{|t|}$ suppressed near the origin. The simplest example with

$$V(r) = r^2 + a^2 r^{-4}, \quad t = -2, \quad (2.14)$$

will be considered below.

The $(2t + 1)$ -diagonal forms of $\mathcal{H}(E)$ may be derived also from some nonpolynomial anharmonicities—a nontrivial example (fractionally anharmonic potentials) may be found in Ref. 12 or 13. The band-matrix form of $\mathcal{H}(E)$ may also be postulated *a priori*, as a phenomenological (possibly, nonlocal) interaction model.¹³

III. ASYMPTOTICALLY SMOOTH CHAIN MODELS

A. Large model space and the AHO example

Notice that the k -dependence of A_k and $B_k = C_{k+1}^T$ in (2.2) is often smooth. In general, the $k \gg 1$ asymptotic AHO estimates

$$\begin{aligned} A_k &= \lambda k^t a(t) (1 + O(1/k)), \\ B_k &= C_{k+1}^T = \lambda k^t b(t) (1 + O(1/k)) \end{aligned} \quad (3.1)$$

$$a_{ij}(t) = \begin{pmatrix} 2t \\ t+i-j \end{pmatrix}, \quad b_{ij}(t) = \begin{pmatrix} 2t \\ i-j \end{pmatrix},$$

$$i, j = 1, 2, \dots, t, \quad t \geq 2,$$

contain the k -dependence in the common scalar factor only. Thus, putting $G_{N+1-k} = \lambda N^t g_k$, the nonlinear matrix mapping $G_{N+1-k} \rightarrow G_{N-k}$ may be approximated by its leading-order form

$$g_{k+1} = a - b(1/g_k) b^T, \quad g_1 = a. \quad (3.2)$$

As a consequence, the AHO MCF expansion

$$G_k = A_k - B_k \{A_{k+1} - B_{k+1} \times [1/(A_{k+2} - \dots)] B_{k+1}^T\}^{-1} B_k^T \quad (3.3)$$

of the effective Hamiltonian (2.4) in a large model space may be replaced by its asymptotic modification $G_k^{(A)} = \lambda k^t g_\infty$, where

$$g_\infty = a - b \{a - b [1/(a - \dots)] b^T\}^{-1} b^T \quad (3.4)$$

is a new, simplified MCF expansion with constant coefficients. This leads to results of the following type.

Theorem 1: In the asymptotically leading-order approximation, the effective $t = 2$ AHO Hamiltonians have the form

$$\mathcal{H}_{(M)}^{(A)} = \begin{pmatrix} A_0 & B_0 & & & \\ & & \ddots & & \\ & & & C_{M-1} & A_{M-1} & B_{M-1} \\ & & & & C_M & G_M^{(A)} \end{pmatrix}, \quad M \gg 1, \quad (3.5)$$

where $M_0 = t$ and

$$G_M^{(A)} = \lambda M^2 \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} + \tilde{\lambda} M \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + O(M), \quad (3.6)$$

$$1 \ll \tilde{\lambda} \ll M.$$

Proof: Formula (3.4) is reducible to the quadratic equation

$$g_\infty = a - b g_\infty^{-1} b^T, \quad (3.7)$$

for $(t \times t)$ -dimensional matrices g_∞ . It defines the stationary value (fixed point, FP) of the mapping $g_k \rightarrow g_{k+1}$. One of the fixed points g_∞ must become the point of accumulation of

the sequence g_{k+1} $k \gg 1$ based on the physical (variational) initialization $g_1 = a$.

For $t = 2$, Eq. (3.7) may be solved easily and its solution is unique (Appendix A). The form of the correction term follows from Appendix B. Q.E.D.

An interesting feature of the preceding result is its independence of the MCF convergence. Although its proof may also be delivered (for g_k and $t = 2$, see Appendix B), the semistable character of the fixed point g_∞ indicates a possibility of divergence of the MCF matrices G_k , caused by the second-order corrections. This is the main short-coming and limitation of the original MCF method.^{9,14}

B. Fixed-point approximation, its sign ambiguity, and perturbation-variational reinterpretation

For the general band-matrix Hamiltonian, we may choose $M \gg 1$ and expect that the effective Hamiltonian $\mathcal{H}_{(M)}^{\text{eff}}(E)$ is again a slowly varying function of the parameter M . In particular, if $A_k \sim A_{k+1}$ and $B_k = C_{k+1}^T \sim B_{k+1}$ in a long interval of indices $k \gg 1$, the fairly reliable fixed-point (FP) approximation F_k to $G_k \sim G_{k+1}$ may be generated along the same lines as in the AHO example, namely, via the quadratic equations of the type (3.7),

$$F_k = A_k - B_k(1/F_k)C_{k+1}. \quad (3.8)$$

These t^2 coupled polynomial equations for the t^2 matrix elements of the approximate effective Hamiltonian $\mathcal{H}_{(k)}^{\text{eff}}$ may be given also the symmetrized form

$$\begin{aligned} F_k \left(\frac{1}{B_k} + \frac{1}{C_{k+1}} \right) F_k - F_k \frac{1}{B_k} A_k \\ - A_k \frac{1}{C_{k+1}} F_k + B_k + C_{k+1} = 0, \end{aligned} \quad (3.9)$$

and solved in accord with the general method described here in Appendix C.

The algebraic root of the quadratic equation (3.9) need not be symmetric and contains an arbitrary pseudo-orthogonal $(t \times t)$ -dimensional matrix S . Vice versa, the symmetric root $F = F^T$ contains only the discrete sign-ambiguity $S(t)$. Its removal is to be based on the MCF origin of the method in a way illustrated on the $t = 1$ example in Ref. 12 [$S(t) = \pm 1$] and on the simplest nontrivial $t = 2$ example with $S(t) = (\pm 1, \pm 1)$ in Appendices A and B. For $t \gg 1$, similar constructions become rather complicated—it is simpler to return to the original MCF iteration itself.¹⁵

For the reasonable values of t , an appropriate elimination of ambiguities will not be difficult whenever we succeed in finding an analog of (3.6) and dominant parameter $\tilde{\lambda}$ simulating the MCF origin of $F_k \sim G_k$. Of course, such an analysis depends on $\mathcal{H}(E)$. For a numerical illustration, we have chosen here the particular example (2.14) with $a = 0$, $t = 2$. The following results were obtained.

(1) The proof of Theorem 2 remains essentially the same. The parameter $\tilde{\lambda}$ may be treated as a variable accelerating the convergence in a trial and error way.

(2) The numerical test (Table I) shows that the practical acceleration of convergence is achieved in a large interval of $\tilde{\lambda} \in (-1, +\infty)$. Similar phenomena may be expected also in the more general cases.

TABLE I. Energy convergence for the various effective interaction corrections [for the harmonic-oscillator potential (2.14) with $a = 0$ and $t = 2$].

$\tilde{\lambda}$	-1.55 ^d	-1.50 ^d	-1.00 ^c	0.00 ^b	$\pm \infty^a$	-2.00 ^c
N						
10	2.542	2.711	3.143	3.291	3.422	
20	2.576	2.742	3.099	3.205	3.293	
30	2.629	2.775	3.080	3.167	3.237	
40	2.671	2.799	3.069	3.144	3.204	
50	2.703	2.818	3.061	3.129	3.182	
100	2.790	2.869	3.043	3.091	3.128	3.505
200	2.854	2.907	3.030	3.064	3.090	3.316
300	2.883	2.924	3.024	3.052	3.073	3.248
400	2.899	2.935	3.021	3.045	3.063	3.210
500	2.910	2.942	3.019	3.040	3.057	3.185

^a $(N \times N)$ -dimensional truncation.

^b Leading-order effective Hamiltonian.

^c Trial acceleration of convergence.

^d The "antivariational" energy estimates.

^e Bad choice—deceleration of convergence.

(3) There exists an interval $[\tilde{\lambda} \in (-1.55, -1.50)]$ such that the corresponding energies converge to the exact value $E = 3$ from below with the increasing cutoff parameter N . Such an "antivariational" phenomenon seems also quite general—it was observed in the $t = 1$ case in Ref. 12. As a perturbation-variational algorithm, it would give the best results, but its rigorous foundation (proof of convergence) as well as practical implementation (location of the corresponding interval of $\tilde{\lambda}$) seems an open problem at present.

IV. FIXED-POINT EXPANSION OF THE EXACT EFFECTIVE HAMILTONIAN

Higher-order corrections to the FP approximation $G_k \sim F_k^{(0)}$ need not always be negligible in the applications. An occurrence of large matrix elements in the corrections $G_k - F_k$ may indicate that the band matrix $\mathcal{H}(E)$ is not smooth enough outside of the chosen model space. Then, we must consider the new quantities defined by the subtraction of $F_k = F_k^{(0)}$,

$$G_k^{(1)} = G_k - F_k^{(0)}, \quad (4.1)$$

and treat them in a recurrent way again.

In analogy with the preceding section, the specification of mapping $G_{k+1}^{(1)} \rightarrow G_k^{(1)}$ [combination of Eqs. (2.5), (3.8), and (4.1)] and of its fixed points $F_k^{(1)}$ (common values of $G_{k+1}^{(1)}$ and $G_k^{(1)}$) may be repeated. This leads to the higher and higher corrections

$$G_k^{(i)} = G_k - F_k^{(0)} - F_k^{(1)} - \dots - F_k^{(i-1)}, \quad (4.2)$$

mappings

$$G_k^{(i)} = A_k^{(i)} - B_k [1/(G_{k+1}^{(i)} + D_k^{(i)})] C_{k+1}, \quad (4.3)$$

$$A_k^{(i)} = A_k - D_k^{(i)} k_{-1},$$

$$D_k^{(i)} = F_{k+1}^{(0)} + F_{k+1}^{(1)} + \dots + F_{k+1}^{(i-1)},$$

and definitions

$$F_k^{(i)} = A_k^{(i)} - B_k [1/(F_k^{(i)} + D_k^{(i)})] C_{k+1} \quad (4.4)$$

of the fixed points with $i = 1, 2, \dots$. For the reasonably smooth matrices $\mathcal{H}$, we may expect that such a scheme is

convergent, $G_k^{(i)} \rightarrow 0$ for $i \rightarrow \infty$, in full analogy with the $t = 1$ example as investigated thoroughly in Ref. 10.

We may symmetrize Eq. (4.4) (cf. the $i = 0$ discussion). Without any difficulties, we may omit the lower as well as upper indices and get the equation

$$\begin{aligned} F(B^{-1} + C^{-1})F + F(C^{-1}D - B^{-1}A) \\ + (DB^{-1} - AC^{-1})F \\ = DB^{-1}A + AC^{-1}D - B - C, \quad B = C^T, \quad i \geq 0. \end{aligned} \quad (4.5)$$

In accord with Appendix C, we may generate its algebraic solution whenever t is reasonably small.

For $i = 0$, the solution of (4.5) or (3.9) was made unique in the way indicated in Appendix B. Indeed, via a geometric interpretation of the mapping $G_{k+1} \rightarrow G_k$ in a vicinity of its fixed points $F_k = F_k[S(t)]$, $S(t) = (\pm 1, \dots, \pm 1)$, we may always specify the physical choice of $S(t)$ and $F_k^{\text{phys}(0)}$ as corresponding to the variational initialization $G_N = A_N$, $N \gg 1$ of the auxiliary sequence G_k .

A priori, the unique specification of the $i > 0$ fixed points $F_k^{\text{phys}(i)}$ also should be easy since their matrix elements may be expected to be small. More rigorously, this expectation is supported not only by the AHO-type examples, but also by the general structure of the method. When we rewrite (4.5) in the form

$$F_k^{(i)} = A_k^{(i)} \frac{1}{B_k^T} (\Delta_k^{(i)} + F_k^{(i)}) \frac{1}{F_k^{(i)} + D_k^{(i)}} B_k^T, \quad i \geq 1, \quad (4.6)$$

where $\Delta_k^{(i)} = F_{k+1}^{(i-1)} - F_k^{(i-1)}$, or

$$\begin{aligned} F(B^{-1} + C^{-1})F + F(C^{-1}D - B^{-1}A) \\ + (DB^{-1} - AC^{-1})F \\ = AC^{-1}\Delta + \Delta B^{-1}A, \quad i \geq 1, \end{aligned} \quad (4.7)$$

we see that we may always choose the sets of signs $S(t) = S^{(i)}(t)$, $i \geq 1$, in such a way that the resulting matrix $F^{(i)}$ is "minimal," $\|\Delta_k^{(i)}\| = O(\|F_k^{\text{phys}(i)}\|^2)$.

Obviously, the norm of Δ measures the "smoothness" of $\mathcal{H}(E)$ and determines the rate of convergence of our final formula, namely, of the FP expansion

$$G_k = \sum_{i=0}^{\infty} F_k^{\text{phys}(i)}. \quad (4.8)$$

Thus, we may conclude that this expansion will converge very quickly whenever the k -dependence of the submatrices A_k , B_k , and C_{k+1} happens to be monotonic or at least sufficiently "smooth." In all such cases we get an estimate

$$\|\Delta_k^{(i)}\| \sim (1/k) \|\Delta_k^{(i-1)}\|, \quad k \gg 1, \quad (4.9)$$

exhibiting a perturbation-type character of convergence in the dimension parameter $1/k$. Our general construction—diagonalization of the band-matrix Hamiltonians—is completed.

When we insert (4.8) into (2.4) with $k = M \gg 1$, we may easily generate the energies as well as the projections of wave functions. The details of this algebraic exercise may be found in Ref. 9.

V. APPLICATIONS

A. The simplest illustration-tridiagonal HO Hamiltonian

Theorem 2: For $i = 0$, the physical HO FP approximant to G_k reads

$$F_k^{(0)} = \frac{1}{2}[(2 + \lambda)a_k - E + D_k], \quad k > k_0(E),$$

$$D_k = \{[(2 + \lambda)a_k - E]^2 - 4\lambda^2 b_k^2\}^{1/2} > 0, \quad \lambda > -1.$$

Proof: The $i = 0$ quadratic equation (4.5) has two roots $F_{k(\pm)}^{(0)} = (1 + \lambda/2)a_k - E/2 \pm D_k/2$. They are both real for $k > k_0(E)$ since

$$\begin{aligned} D_k^2 &= (2 + \lambda)^2 a_k^2 + O(k) - 4\lambda^2 k^2 \\ &= 16(\lambda + 1)k^2 + O(k), \quad k \gg 1. \end{aligned}$$

One of them is to be eliminated as unphysical.

In the $k \gg 1$ asymptotic region the mapping $G_{k+1} \rightarrow G_k$ with $G_k = \lambda k \gamma_k$ has the asymptotic form

$$\gamma_k = 2 + 4\lambda^{-1} - 1/\gamma_{k+1} + O(1/k), \quad k \gg 1.$$

Its geometric interpretation shows that this mapping is contractive for almost all initializations including $\gamma_N = A_N/\lambda N$ (Fig. 1). This implies that after a sufficient number of iterations, the sequence γ_k accumulates near the stable fixed point

$$\gamma_k \sim \gamma_k^{(+)} = 1 + 2\lambda^{-1}(1 + (1 + \lambda)^{1/2}), \quad \lambda > -1$$

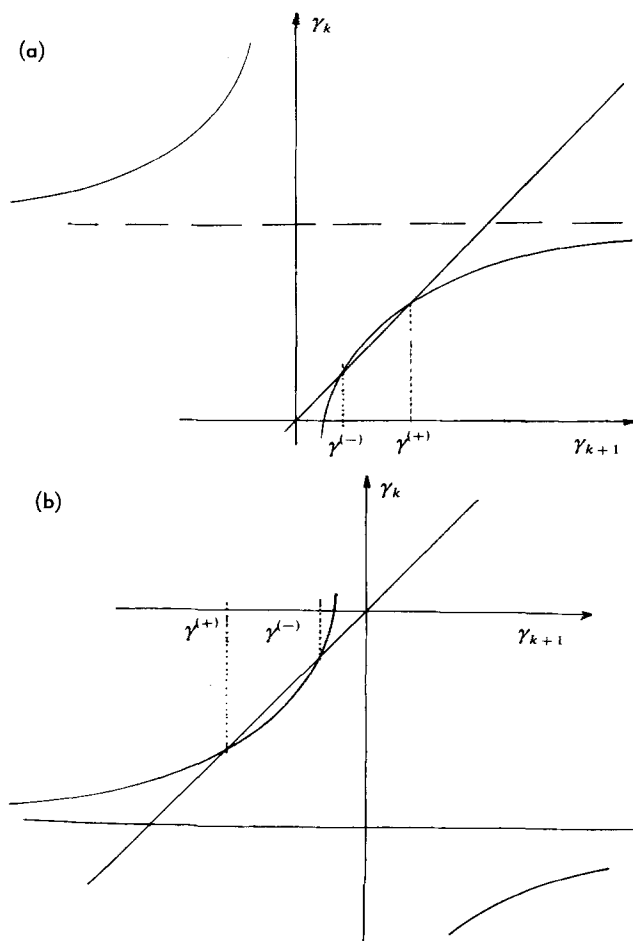


FIG. 1. Geometric interpretation of the HO mapping $G_{k+1} \rightarrow G_k$. (a) $\lambda = 1/\sinh^2 x > 0$, $\gamma_k^{(\pm)} = \exp(\pm 2x)$. (b) $\lambda = -1/\cosh^2 x > -1$, $\gamma_k^{(\pm)} = -\exp(\pm 2x)$.

(see Fig. 1 again). This establishes the convergence of the MCF expansion (2.12) and eliminates the $\gamma^{(-)}$ root (sign ambiguity) on the variational grounds. Q.E.D.

Alternative proof of theorem 2: We may omit the variational argument and start from the explicit solution

$$G_0 = 0, \quad X_0 \neq 0$$

of the model-space condition (2.8) with $t = M_0 = 1$. When interpreted as a pair of initializations of the recurrences (2.5) and (2.7), respectively, it fixes the formal projections $\langle n | \psi \rangle$. The independent physical requirement

$$\|\psi\| < \infty$$

should fix the binding energies. Among the two roots $\gamma^{(\pm)}$ of the $i = 0$ equation (4.5), we have to pick out the one compatible with the convergence requirement

$$|C_k/G_k| = |1/\gamma_k| < 1, \quad k \gg 1$$

which eliminates the non-MCF root $\gamma^{(-)}$ as spurious again. Q.E.D.

In the $i = 1$ approximation it is sufficient to evaluate the quantities

$$\begin{aligned} \Delta_k^{(1)} &= F_{k+1}^{(0)} - F_k^{(0)} = 2 + \lambda + [4(2 + \lambda)^2(2k + l + \frac{1}{2}) \\ &\quad - 4E(2 + \lambda) - 4\lambda^2(2k + l + \frac{1}{2})]/(D_{k+1} + D_k) \\ &= 2 + \lambda + 4(1 + \lambda)^{1/2} + O(1/k), \end{aligned}$$

and find the $i = 1$ solution of (4.5). After the appropriate insertions, we obtain the result

$$\begin{aligned} F_{k(\pm)}^{(1)} &= +2k(1 + \lambda)^{1/2}[-1 \pm (1 + O(1/k))], \\ F_k^{(1)\text{phys}} &= F_{k(+)}^{(1)} = O(1), \quad k \gg 1. \end{aligned}$$

It is compatible with the smallness requirement $F^{\text{phys}} \sim \Delta^{1/2}$ and does not increase with the increasing model space. The next corrections

$$\Delta_k^{(n+1)} = O(1/k^n), \quad n = 1, 2, \dots,$$

become negligible for the large dimensions of the model space.

Our FP expansion converges as a geometric series in the HO example. This feature will be preserved also in the more complicated systems of a similar character, in spite of more complicated technicalities. For $t = 2$, all the related discussion preserves even a non-numerical character—this will be proved in the following final section.

B. Pentadiagonal Hamiltonians and their diagonalization by algebraic means

Theorem 3: A Schrödinger-type equation (2.1) with the pentadiagonal “Hamiltonian” $\mathcal{H}(E)$ may be represented by its finite-dimensional equivalents (2.3) + (2.4) with

$$G_k = Z^{-1} U \left[\sum_{i=0}^{\infty} \begin{pmatrix} x^{(i)} & y^{(i)} \\ y^{(i)} & z^{(i)} \end{pmatrix} \right] U Z^{-1}, \quad k \gg 1, \quad (5.1)$$

where the diagonal Z and unitary matrix $U = U^T$, as well as the matrix elements $x^{(i)}$, $y^{(i)}$, and $z^{(i)}$ may be all defined by the elementary algebraic expressions.

Proof: First, let us notice that all the quadratic equa-

tions (4.5) contain the common i -independent factor $B^{-1} + C^{-1}$ ($\equiv$ matrix A of Appendix C). It is convenient to write $B = C^T$ and $1/B + 1/B^T = ZRZ$ with the arbitrary diagonal matrix Z and real and symmetric R . Since the secular equation $\det(R - E) = 0$ (ordinary quadratic equation) admits the algebraic solution, we may write

$$R = \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} E_+ & \\ & E_- \end{pmatrix} \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix}.$$

To avoid the transpositions, we may insert here the identity matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and get

$$B^{-1} + C^{-1} = ZU \begin{pmatrix} E_- & \\ & E_+ \end{pmatrix} UZ, \quad (5.2)$$

$$U = \begin{pmatrix} \sin \varphi & \cos \varphi \\ \cos \varphi & -\sin \varphi \end{pmatrix}.$$

Using (5.2) and denoting

$$F = Z^{-1}U \begin{pmatrix} x & y \\ y & z \end{pmatrix} UZ^{-1}, \quad (5.3)$$

we may simplify the structure of Eq. (4.5) when we insert $I = ZZ^{-1}$ and $I = UU$ whenever necessary. With respect to the possible changes of signs of $E_{\pm}$, we have to distinguish between the four cases, with $E_- = \pm g_1^2$ and $E_+ = \pm g_2^2$. For the sake of simplicity, we shall consider here only one of them, say, $E_- = -g_1^2$ and $E_+ = +g_2^2$, which gives

$$\begin{pmatrix} x & y \\ y & z \end{pmatrix} \begin{pmatrix} -g_1^2 & 0 \\ 0 & g_2^2 \end{pmatrix} \begin{pmatrix} x & y \\ y & z \end{pmatrix} + \begin{pmatrix} x & y \\ y & z \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} x & y \\ y & z \end{pmatrix} + \begin{pmatrix} h_1 & h_2 \\ h_2 & h_3 \end{pmatrix} = 0, \quad (5.4)$$

in place of (4.5). In the AHO case, the coefficients are real (Appendix D).

The latter equation has obviously the form of three independent scalar relations

$$\begin{aligned} -g_1^2 x^2 + 2ax + g_2^2 y^2 + 2cy + h_1 &= 0, \\ -g_1^2 y^2 + 2by + g_2^2 z^2 + 2dz + h_3 &= 0, \\ -g_1^2 xy + bx + g_2^2 yz + cz + (a+d)y + h_2 &= 0. \end{aligned} \quad (5.5)$$

When we denote

$$\begin{aligned} h_4 &= h_1 + (a/g_1)^2 - (c/g_2)^2, \\ h_5 &= h_3 + (b/g_1)^2 - (d/g_2)^2, \\ h_6 &= h_2 + (ab/g_1^2) - (cd/g_2^2), \end{aligned}$$

and, in analogy with Appendix A,

$$\begin{aligned} \alpha + \beta &= g_1 x - a/g, & \alpha - \beta &= g_2 y + c/g_2, \\ \gamma + \delta &= g_1 y - b/g_1, & \gamma - \delta &= g_2 z + d/g_2, \end{aligned} \quad (5.6)$$

we may distinguish again the various sign combinations. Omitting this discussion here, we obtain the two definitions

$$\beta = h_4/4\alpha, \quad \delta = h_5/4\gamma$$

accompanied by the third ordinary quadratic equation

$$\alpha\delta + \beta\gamma = h_6/2,$$

i.e., an equation analogous to (A5),

$$h_5 \rho + h_4/\rho = 2h_6, \quad \rho = \alpha/\gamma,$$

which defines the two values of the ratio

$$\rho = \rho_{\pm} = (1/h_5)[h_6 \pm (h_6^2 - h_4 h_5)^{1/2}]. \quad (5.7)$$

Finally, an analog of (A6),

$$\begin{aligned} (g_2 - g_1 \rho_{\eta}) \gamma^2 + (bg_2/g_1 + cg_1/g_2) \gamma \\ + \frac{1}{4}(g_2 h_5 + g_1 h_4/\rho_{\eta}) = 0, \quad \eta = \pm 1, \end{aligned}$$

represents a compatibility of the two independent definitions of γ in (5.6) and defines the four values of γ :

$$\begin{aligned} \gamma = \gamma_{\epsilon\eta} = \frac{1}{2(g_2 - g_1 \rho_{\eta})} \left(-\frac{bg_2}{g_1} - \frac{ag_1}{g_2} + \epsilon[a^2 + d^2 \right. \\ \left. + 2bc + g_1^2 h_1 - g_2^2 h_3 + 2g_1 g_2 \eta (h_6^2 - h_4 h_5)^{1/2}]^{1/2} \right), \\ \epsilon, \eta = \pm 1. \end{aligned} \quad (5.8)$$

The backward insertions are then trivial and give

$$z = z_{\epsilon\eta} = -d/g_2^2 + \gamma_{\epsilon\eta}/g_2 - h_5/4g_2\gamma_{\epsilon\eta}, \quad (5.9)$$

etc. We may summarize that the two-dimensional quadratic equation (4.5) is completely reducible to the sequence of a few one-dimensional ones. Q.E.D.

The algebraic construction in the preceding proof is straightforward and elementary. It is a little bit cumbersome and lengthy in practice so that we may assume that the energy- and coupling-dependence of the FP expansion (5.1) will be generated by an appropriate symbolic-manipulation procedure in most cases. In a large model space, we may use also the asymptotic expansions as a guide to the simplifications.

VI. SUMMARY

We have considered the class of physical models based on the chaining hypothesis¹ and described by the multidagonal Hamiltonians. We have shown that at least some of these models may admit a reasonably simple mathematical treatment. Our conclusions are of the following three types.

(1) As a reformulation of the relevant physical information (interaction between the neighboring basis states only), our FP method is an alternative to the standard perturbation theories. As a guide how to construct the effective Hamiltonians in a systematic way, it may even be given sometimes the direct perturbative interpretation. In our anharmonic oscillator example, the small parameter characterizing the convergence of our FP series may be identified with the inverse dimension $1/d$.

(2) As a somewhat nonstandard algebraic formalism the present method leads to the new explicit, recurrent, or asymptotic relations for the wave function and/or for the effective interaction. It is based on the special block-tridiagonal structure of the Hamiltonian and on its geometric fixed-point reinterpretation. This may specify, e.g., the limitations and/or possible extensions of the classical perturbative expansions. Furthermore, in contrast to the standard numerical or continued fractional methods (and their results equivalent to the present $i = 0$ approximation), the corrections to the effective Hamiltonian have an additive character. This is another analogy with the perturbation theory.

(3) As a practical numerical or seminumerical algorithm, the present FP approach enables one to save significantly the computer time. Indeed, besides the acceleration of convergence of the ordinary truncation eigenvalue method, the complementary “antivariational” lower bounds for energies were obtained in the illustrative example. The existence of such a “sandwiching of eigenvalues” seems a more general phenomenon¹² to be investigated more thoroughly in the future.

APPENDIX A: ASYMPTOTIC UNIQUENESS OF THE AHO FP APPROXIMANT

Lemma 1: The fixed point of the $t = 2$ mapping (3.2) is unique.

Proof: The mapping $g_k (= 1/f_k) \rightarrow g_{k+1} (= 1/f_{k+1})$, i.e.,

$$f_{k+1} = \left[\begin{pmatrix} 6 & 4 \\ 4 & 6 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix} f_k \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \right]^{-1} \quad (\text{A1})$$

is assumed to have a fixed point $f = f_k = f_{k+1}$. Putting

$$f = U\eta U^T, \quad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \eta = \begin{pmatrix} A & B \\ B & D \end{pmatrix},$$

we obtain its symmetrized ($f = f^T$) algebraic definition

$$\eta \begin{pmatrix} 6 & 0 \\ 0 & -2 \end{pmatrix} \eta + \eta \begin{pmatrix} 10 & -20 \\ 4 & -6 \end{pmatrix} + \begin{pmatrix} 10 & 4 \\ -20 & -6 \end{pmatrix} \eta + \begin{pmatrix} -2 & 0 \\ 0 & 6 \end{pmatrix} = 0,$$

which is equivalent to the three coupled scalar quadratic equations

$$\begin{aligned} (3A + 5)^2 &= 3(B - 2)^2 + 16, \\ (3B - 10)^2 &= 3(D + 3)^2 + 64, \\ (3A + 5)(3B - 10) &= 3(B - 2)(D + 3) - 32. \end{aligned} \quad (\text{A2})$$

In terms of the linear combinations of the four new (primed) variables

$$\begin{aligned} 3A + 5 &= \epsilon_1(A' + B'), \quad \sqrt{3}(B - 2) = \epsilon_2(A' - B'), \\ 3B - 10 &= \epsilon_3(D' + C'), \quad \sqrt{3}(D + 3) = \epsilon_4(D' - C'), \\ \epsilon_i &= \pm 1, \quad i = 1, 2, 3, 4, \end{aligned} \quad (\text{A3})$$

we may reduce the first two items in (A2) to the definitions

$$B' = 4/A', \quad C' = 16/D'. \quad (\text{A4})$$

Considering the third condition in (A2) with

$$\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon_4 = +1$$

for the time being, we get

$$2A'/D' + D'/2A' = -2, \quad (\text{A5})$$

with the unique and doubly degenerate real solution $D' = -2A'$.

When we compare the second and the third item in (A3), we see that the original parameter B is defined in two ways now. Their compatibility requires

$$\sqrt{3}(A' - 4/A') = 4 - (2A' + 8/A'), \quad (\text{A6})$$

and fixes the remaining freedom—this is again a quadratic

equation with the degenerate solution. Hence, we have

$$A' = 4 - 2\sqrt{3}, \quad B' = 4 + 2\sqrt{3}, \quad (\text{A7})$$

$$D' = -8 + 4\sqrt{3}, \quad C' = -8 - 4\sqrt{3}.$$

Now, let us vary the signs ϵ_2 (by an interchange of A' and B' , i.e., $A' \leftrightarrow B'$), ϵ_4 (interchanging $D' \leftrightarrow C'$), ϵ_1 (by the transformation $A' \rightarrow -B'$, $B' \rightarrow -A'$), or ϵ_3 (changing $D' \rightarrow -C'$ and $C' \rightarrow -D'$). Obviously, we get

$$A' = 4\epsilon_1 - 2\epsilon_2\sqrt{3}, \quad B' = 4\epsilon_1 + 2\epsilon_2\sqrt{3}, \quad (\text{A8})$$

$$C' = -8\epsilon_3 - 4\epsilon_4\sqrt{3}, \quad D' = -8\epsilon_3 + 4\epsilon_4\sqrt{3}.$$

Finally, the insertion into (A2) implies that $3A + 5 = 8\epsilon_1^2 = 8$, etc., so that the resulting solution is unique,

$$A = 1, \quad B = -2, \quad D = 5. \quad (\text{A9})$$

Q.E.D.

APPENDIX B: MCF CONVERGENCE IN THE AHO-TYPE CASES

Lemma 2: The matrix continued fraction

$$f = \left\{ \begin{pmatrix} 6 & 4 \\ 4 & 6 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix} \left[\begin{pmatrix} 6 & 4 \\ 4 & 6 \end{pmatrix} - \dots \right]^{-1} \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \right\}^{-1} \quad (\text{B1})$$

is convergent and equal to

$$\begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}.$$

Its finite approximants have the form

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \left[\begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} \alpha & -\xi \\ -\xi & \gamma \end{pmatrix} \right]^{-1} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (\text{B2})$$

with $5 > \alpha > \xi > \gamma > 0$.

Proof: Equation (B1) or (3.2) may be given the form

$$\begin{aligned} \tilde{\varphi} &= \begin{pmatrix} 10 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} \varphi^{-1} \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}, \\ \tilde{\varphi} &= U g_{k+1} U = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} \tilde{\alpha} & -\tilde{\xi} \\ -\tilde{\xi} & \tilde{\gamma} \end{pmatrix}, \end{aligned} \quad (\text{B3})$$

$$\varphi = U g_k U = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} \alpha & -\xi \\ -\xi & \gamma \end{pmatrix}.$$

This coincides with the rational mapping

$$\begin{aligned} \tilde{\alpha} &= (\alpha + 8\xi + 16\gamma + 5\rho)/(1 + \epsilon), \\ \tilde{\xi} &= (\xi + 4\gamma + 2\rho)/(1 + \epsilon), \\ \tilde{\gamma} &= (\gamma + \rho)/(1 + \epsilon), \\ \tilde{\rho} &= \rho/(1 + \epsilon), \\ \tilde{\epsilon} &= (\alpha + 12\xi + 37\gamma + 19\rho)/(1 + \epsilon), \end{aligned} \quad (\text{B4})$$

with the two additional auxiliary variables

$$\rho = \alpha\gamma - \xi^2 = \det \left(\varphi - \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} \right), \quad \epsilon = \alpha + 4\xi + 5\gamma + \rho.$$

All the relations (B4) have the same form

$$\tilde{x} = (x + v_x)/(U_x x + V_x + 1), \quad (\text{B5})$$

with

$$\begin{aligned}v_\alpha &= 8\xi + 16\gamma + 5\rho, & v_\xi &= 4\gamma + \rho, & v_\gamma &= 0 = v_\rho, \\v_\epsilon &= 8\xi + 32\gamma + 18\rho, \\U_\alpha &= 1, & U_\xi &= 4, & U_\gamma &= 5, & U_\rho &= 1 = U_\epsilon, \\V_\alpha &= 4\xi + 5\gamma + \rho, & V_\xi &= \alpha + 5\gamma + \rho, \\V_\rho &= \alpha + 4\xi + 5\gamma, \\V_\gamma &= \alpha + 4\xi + \rho, & V_\epsilon &= 0.\end{aligned}$$

They all possess also a geometric interpretation similar to that given in Fig. 1. We may observe the following.

- (i) $\alpha = \xi = \gamma = 0$ is a fixed point of the mapping (B3).
- (ii) The numerical test (Table II) confirms the convergence to the zero fixed point. In particular, the trivial initialization of the original quantities f is reflected here by the initial choice of a "correction" $\alpha = 5$, $\xi = 2$, and $\gamma = 1$.
- (iii) From any initial set of the positive values α , ξ , and γ such that also $\rho > 0$, we get $\epsilon > 0$ and $\tilde{\alpha} > 0$, $\tilde{\xi} > 0$, $\tilde{\gamma} > 0$ and $\tilde{\rho} > 0$: The positivity of parameters is preserved by the mapping [Eq. (B4)].
- (iv) We see from Fig. 2 that each sequence $x = (\alpha, \xi, \gamma, \rho, \text{ or } \epsilon)$ should independently accumulate near its stable fixed point

$$x_{(+)} = [-V_x + (V_x^2 + 2U_x v_x)^{1/2}] / 2U_x, \quad (\text{B6})$$

the value of which depends on the remaining parameters.

- (v) In more detail, Eqs. (B3) and (B6) imply the following.

- (a) The sequence of ρ 's is monotonic and decreases to zero.
- (b) For a sufficiently small ρ , we may partially "linearize" Eq. (B3) by omitting ρ 's and see that γ starts to approach $\gamma_{(+)}$ in a monotonic way [Fig. 2(a)].
- (c) Similar procedure ("linearization") may be then repeated for ξ and also for α or ϵ . From Eq. (B6) we see that the stationary values $x_{(+)}$ become strictly ordered as soon as $\rho \ll 1$. We get

$$\begin{aligned}\gamma \sim \gamma_{(+)} &\sim [-V_\gamma + (V_\gamma^2 + 20\rho)^{1/2}] / 10 \gtrsim (\rho/5)^{1/2} \gg \rho, \\ \xi \sim \xi_{(+)} &\sim [-V_\xi + (V_\xi^2 + 64\gamma)^{1/2}] / 8 \gtrsim \sqrt{\gamma} \gg \gamma, \\ \alpha \sim \alpha_{(+)} &[-V_\alpha + (V_\alpha^2 + 4v_\alpha)^{1/2}] / 2 \gtrsim (8\xi)^{1/2} \gg \xi.\end{aligned} \quad (\text{B7})$$

- (d) The same type of ordering takes place from the very beginning, even for large ρ 's (cf. Table II); and vice versa, if the ordering of the deviations is violated by our choice of another initialization, the overall convergence would be slowed down. Hence, the nonlinearity of the original relations (B3) is essential and a rate of the overall conver-

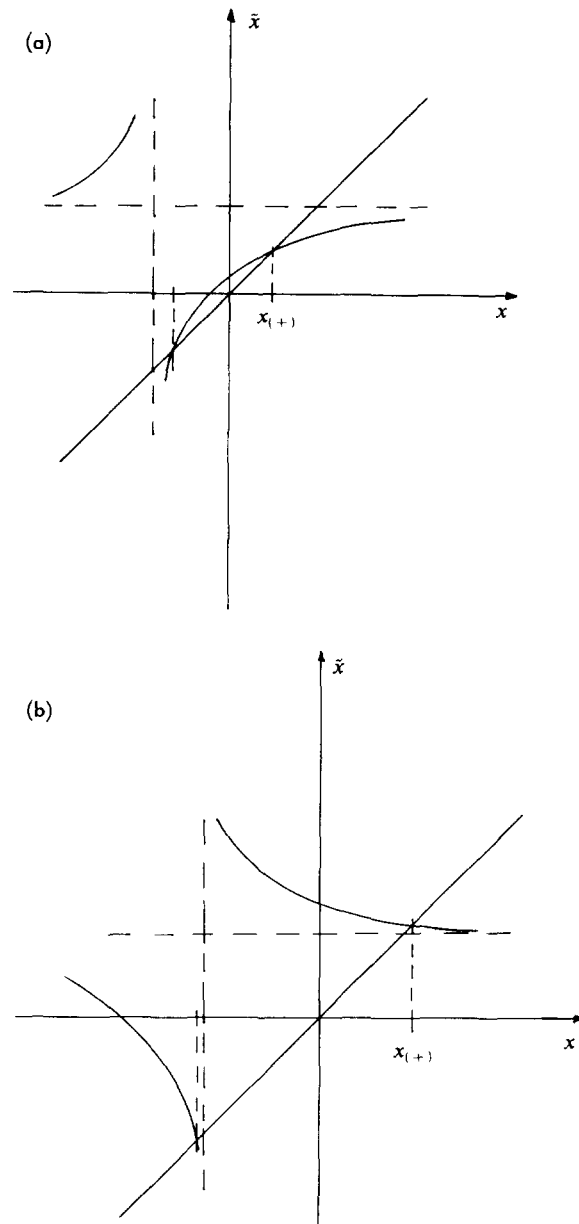


FIG. 2. Two possible patterns of convergence $x \rightarrow x_{(+)}$ in Appendix B. (a) $v < (V+1)/U$. (b) $v > (V+1)/U$.

gence is controlled by the convergence of α 's or ϵ 's,

$$\tilde{\alpha} = \alpha / (1 + \alpha), \quad \tilde{\epsilon} = \epsilon / (1 + \epsilon), \quad (\text{B8})$$

in the simplest approximation. This completes the proof.

TABLE II. Sample of the MCF convergence [Eq. (B4)].

M	γ_M	ξ_M	α_M	M	γ_M	ξ_M	α_M
0	1.000	2.000	5.000	5	0.003 4	0.0419	0.6701
1	0.100	0.400	2.100	10	0.000 5	0.0124	0.3642
2	0.029	0.171	1.362	15	0.000 1	0.0058	0.2501
3	0.012	0.095	1.012	20	0.000 08	0.0034	0.1905
4	0.006	0.061	0.806	25	0.000 04	0.0022	0.1538

APPENDIX C: QUADRATIC EQUATION FOR MATRICES AND ITS SOLUTION

For the $(t \times t)$ -dimensional matrix F , let us consider the equation

$$FAF + \frac{1}{2}(BF + FB^T) + C = 0, \quad (C1)$$

where $A = A^T$, $C = C^T$, and the symmetric root $F = F^T$ is to be determined by the algebraic means.

We shall assume that there exists a matrix D such that $B = 2DA$. Then, obviously, we may write

$$(F + D)A(F + D^T) = DAD^T - C, \quad (C2)$$

which is equivalent to (C1).

Next, the real and symmetric matrices may be diagonalized by some orthogonal matrices U and W ,

$$\begin{aligned} A &= UaU^T, \quad DAD^T - C = WbW^T, \\ a &= \epsilon\alpha^2, \quad b = \eta\beta^2, \\ \epsilon &= \begin{pmatrix} \pm 1 & & \\ & \pm 1 & \\ & & \ddots \end{pmatrix}, \quad \eta = \begin{pmatrix} \pm 1 & & \\ & \pm 1 & \\ & & \ddots \end{pmatrix}. \end{aligned} \quad (C3)$$

Hence, with an arbitrary matrix S which satisfies the pseudo-orthogonality relations

$$SeS^T = \eta, \quad (C4)$$

and may be constructed in accord with the underlying group theory in various ways, we obtain the general solution of (C1) in the form

$$F = -D + W\beta S\alpha^{-1}U^T. \quad (C5)$$

Finally, up to the possible changes of signs, the $t(t-1)/2$ independent matrix elements of S become fixed by the $t(t-1)/2$ symmetry requirements $F = F^T$.

The above construction may be illustrated by the $t = 2$ example of Appendix A once more. We have

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad a = \begin{pmatrix} 6 & 0 \\ 0 & -2 \end{pmatrix}, \quad (C6)$$

$$W = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} \frac{160}{3} & 0 \\ 0 & 0 \end{pmatrix},$$

so that the general solution F has the form (C5),

$$\begin{aligned} F &= \begin{pmatrix} -\frac{2}{3} & 2 \\ \frac{10}{3} & -3 \end{pmatrix} \\ &+ 8 \begin{pmatrix} \rho_1 \cosh \xi / 6, \rho_2 \sinh \xi / 2\sqrt{3} \\ -\rho_1 \cosh \xi / 3, -\rho_2 \sinh \xi / \sqrt{3} \end{pmatrix}, \\ \rho_{1,2} &= \pm 1, \quad \xi \in (0, \infty). \end{aligned} \quad (C7)$$

It is to be complemented by the only symmetry requirement,

$$2\rho_1 \cosh \xi + \rho_2 \sqrt{3} \sinh \xi = 1, \quad (C8)$$

which eliminates the variable ξ .

The solution of (C8) is straightforward and gives

$$e^\xi = 1/(2\rho_1 + \rho_2\sqrt{3}).$$

Hence, the resulting formula $\cosh \xi = 2\rho_1$ implies that $\rho_1 = 1$ and, finally, $\rho_2 = -\operatorname{sgn} \xi$. After the insertion into (C7), we see that $F = F^T$ is unique and equal to the solution obtained in Appendix A.

APPENDIX D: AHO EXAMPLE AND EQ. (5.4)

(i) The AHO $t = 2$ example may be given an explicit form without any computer—using (2.13) together with the definition (5.2) and

$$\begin{aligned} Z &= b_{n+1}^{-1/2} \begin{pmatrix} 1/b_n & 0 \\ 0 & 1/b_{n+2} \end{pmatrix}, \\ R &= \begin{pmatrix} 2b_n & -(a_{n+1} + a_{n+2}) \\ -(a_{n+1} + a_{n+2}) & 2b_{n+2} \end{pmatrix}. \end{aligned}$$

The simple prescriptions $[n = n(k) = 2k + M_0 - 2]$

$$\begin{aligned} E_{\pm} &= b_n + b_{n+2} \\ &\pm [(b_{n+2} - b_n)^2 + (a_{n+1} + a_{n+2})^2]^{1/2}, \\ \tan 2\varphi &= -(a_{n+1} + a_{n+2})/(b_{n+2} - b_{n-1}), \\ &\vdots \end{aligned}$$

specify the explicit AHO form of the FP series (5.1).

(ii) We may easily verify that $E_+ > 0$ and $E_- < 0$ since the corresponding expression

$$(2n + l + \frac{3}{2})^4 - (n + 1)(n + 3)(n + l + \frac{3}{2})(n + l + \frac{7}{2})$$

is a positive and monotonically increasing function of k and l in the whole interval of interest (say, for $k \geq 0$ and $l \geq -\frac{3}{2}$). Hence, g_1 and g_2 of Eq. (5.4) are both real. The value of the angle φ is also a monotonically decreasing function of k and l such that $\pi/4 < \varphi < \pi/2$. On the boundaries, $\tan \varphi = (1 + \sqrt{7})/\sqrt{6}$ for $k = 0$ and $l = -\frac{3}{2}$, and $\varphi \rightarrow \pi/4$ for $k \rightarrow \infty$ and/or $l \rightarrow \infty$.

(iii) The AHO solution admits the alternative forms of denotation, e.g.,

$$\begin{aligned} U &= \sigma_1 \cos \psi + \sigma_2 \sin \psi, \quad \psi = \varphi - \pi/4, \\ \sigma_1 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \sigma_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \\ \sigma_1^2 &= \sigma_2^2 = I, \quad \sigma_1 \sigma_2 = -\sigma_2 \sigma_1. \end{aligned}$$

This will not be discussed here.

¹D. W. Bullett, R. Haydock, V. Heine, and M. J. Kelly, *Solid State Physics*, edited by H. Ehrenreich, F. Seitz, and D. Turnbull (Academic, New York, 1980), Vol. 35, and references contained therein.

²H. Feshbach, A. Kerman, and S. Koonin, *Ann. Phys. (NY)* **125**, 429 (1980).

³S. Graffi and V. Grecchi, *Lett. Nuovo Cimento* **12**, 425 (1975).

⁴M. Znojil, *Phys. Rev. D* **24**, 903 (1981).

⁵I. G. Halliday and P. Suranyi, *Phys. Rev. D* **21**, 1529 (1980).

⁶H. S. Wall, *Analytic Theory of Continued Fractions* (Van Nostrand, New York, 1948).

⁷N. I. Akhiezer, *The Classical Moment Problem* (Oliver and Boyd, London, 1955).

⁸G. A. Baker, *Essential of Padé Approximants* (Academic, New York, 1975).

⁹M. Znojil, *J. Math. Phys.* **21**, 1629 (1980).

¹⁰M. Znojil, *J. Phys. A: Math. Gen.* **13**, 2375 (1980), Appendix.

¹¹H. Feshbach, *Ann. Phys. (NY)* **5**, 357 (1958).

¹²M. Znojil, *J. Phys. A: Math. Gen.* **16**, 279 (1983); 293 (1983).

¹³R. R. Whitehead, A. Watt, G. P. Flessass, and M. A. Nagarajan, *J. Phys. A: Math. Gen.* **15**, 1217 (1982).

¹⁴M. Znojil, *Phys. Rev. D* **26**, 3750 (1982).

¹⁵D. F. Scofield, *Phys. Rev. Lett.* **29**, 811 (1972).